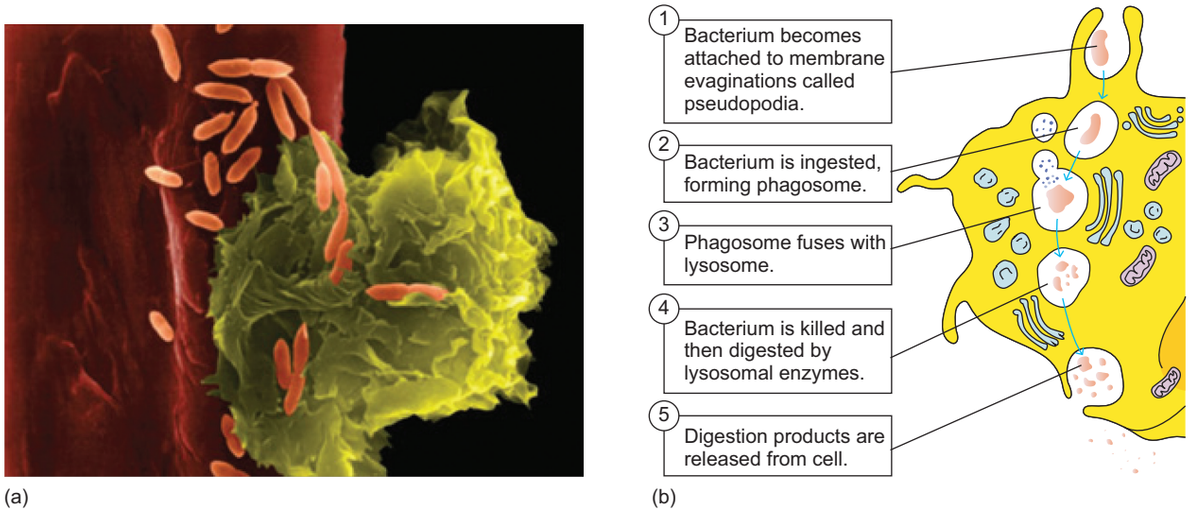


Figure 13.20: (a) Scanning electron micrograph of alveolar macrophage phagocytosis of E. coli bacteria on the outer surface of a blood vessel in the lung pleural cavity. (b) Steps in the phagocytosis of a bacterium.

D. Inflammation

Tissue damage caused by a wound or by an invading pathogenic microorganism induces a complex sequence of events collectively known as inflammatory response. Inflammation can either be acute or chronic. The gross features were described over 2000 years ago and are still known as the cardinal signs of inflammation: redness (rubor), warmth (calor), pain (dolor), swelling (tumor), and loss of function (functiolaesa)

The cardinal signs of inflammation reflect the three major events of an inflammatory response.

1. Vasodilation (an increase in the diameter of blood vessels) of nearby capillaries occurs as the vessels that carry blood away from the affected area constrict. This results in engorgement of the capillary network. The engorged capillaries are responsible for tissue redness (erythema) and an increase in temperature.
2. An increase in capillary permeability facilitates an influx of fluid and cells from the engorged capillaries into the tissue. The fluid that accumulates (exudate) has much higher protein content. Accumulation of exudate contributes to tissue swelling (edema)

Infobits

Reactive Nitrogen Species: Highly cytotoxic antimicrobial compounds formed by the combination of nitric oxide and superoxide anion within phagocytes such as neutrophils and macrophages.

Reactive Oxygen Species (ROS): Highly reactive compounds such as superoxide anion O_2, hydroxyl radicals (OH)(OH^-), hydrogen peroxide (H_2O_2), and hypochlorous acid (HClO) that are formed from oxygen under many conditions in cells and tissues, including microbe-activated innate responses of phagocytic cells; have anti-microbial activity.

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